

LSTMCell#

<https://pytorch.org/docs/stable/generated/torch.nn.modules.rnn.LSTMCell.html>

Description:

A long short-term memory (LSTM) cell. where σ is the sigmoid function, and $\odot$ is the Hadamard product. `input_size` (int) – The number of expected features in the input `hidden_size` (int) – The number of features in the hidden state `h` `bias` (bool) – If False, then the layer does not use bias weights `b_ih` and `b_hh`. Default: True

Function Signature

```
class torch.nn.modules.rnn.LSTMCell(input_size, hidden_size,
bias=True, device=None, dtype=None)[source]#
```

Parameters

Inputs: `input`, `(h_0, c_0)`

Outputs: `(h_1, c_1)`

Variables

Code Examples

```
>>> rnn = nn.LSTMCell(10, 20) # (input_size, hidden_size)
>>> input = torch.randn(2, 3, 10) # (time_steps, batch,
input_size)
>>> hx = torch.randn(3, 20) # (batch, hidden_size)
>>> cx = torch.randn(3, 20)
>>> output = []
>>> for i in range(input.size()[0]):
...     hx, cx = rnn(input[i], (hx, cx))
...     output.append(hx)
>>> output = torch.stack(output, dim=0)
```

Notes & Warnings

NOTE All the weights and biases are initialized from $U(-k, k)\sqrt{k}$ where $k = \frac{1}{\text{hidden_size}}$

Detailed Documentation

Documentation

A long short-term memory (LSTM) cell.

where σ is the sigmoid function, and $\odot$ is the Hadamard product.

`input_size` (int) – The number of expected features in the input x

`hidden_size` (int) – The number of features in the hidden state h

`bias` (bool) – If False, then the layer does not use bias weights b_{ih} and b_{hh} . Default: True

`input` of shape (batch, `input_size`) or (`input_size`): tensor containing input features

`h_0` of shape (batch, `hidden_size`) or (`hidden_size`): tensor containing the initial hidden state

`c_0` of shape (batch, `hidden_size`) or (`hidden_size`): tensor containing the initial cell state

If (`h_0`, `c_0`) is not provided, both `h_0` and `c_0` default to zero.

`h_1` of shape (batch, `hidden_size`) or (`hidden_size`): tensor containing the next hidden state

`c_1` of shape (batch, `hidden_size`) or (`hidden_size`): tensor containing the next cell state

`weight_ih` (torch.Tensor) – the learnable input-hidden weights, of shape ($4 \times \text{hidden_size}$, `input_size`)

`weight_hh` (torch.Tensor) – the learnable hidden-hidden weights, of shape ($4 \times \text{hidden_size}$, `hidden_size`)

`bias_ih` – the learnable input-hidden bias, of shape ($4 \times \text{hidden_size}$)

`bias_hh` – the learnable hidden-hidden bias, of shape ($4 \times \text{hidden_size}$)

All the weights and biases are initialized from $U(-k, k) \sqrt{\frac{1}{\text{hidden_size}}}$ where $k = \frac{1}{\sqrt{\text{hidden_size}}}$

On certain ROCm devices, when using float16 inputs this module will use different precision for backward.